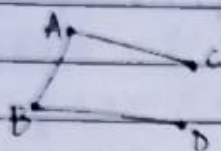


MODULE - 5

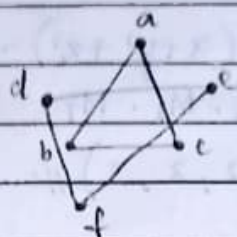
FUNDAMENTALS OF GRAPH

* CONNECTED & DISCONNECTED GRAPHS

A graph G is said to be connected if there is atleast one path b/w every two vertices in G and disconnected if G has atleast one pair of vertices b/w which there is no path.



connected

a b c d e f - components
of the
graph

disconnected

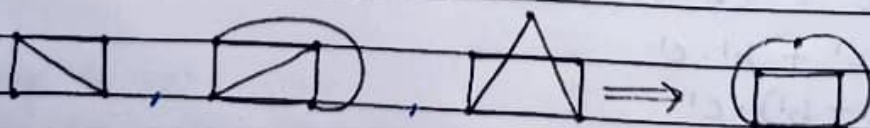
* THEOREM

A simple graph with n vertices & k components can have atmost $\frac{(n-k)(n-k+1)}{2}$ edges.

* PLANAR GRAPHS

A graph G is said to be planar if there exists some geometric rep of G which can be drawn on a plane such that no two of its edges intersect. P.O.I are called crossovers.

eg:



If G can be redrawn to be planar $\rightarrow$ it is planar.

non-planar:



* COMPLETE GRAPH (K)

Each vertex has an edge.

K_3 :



K_5 :



* WALKS, PATHS & CIRCUITS

WALK

PATH

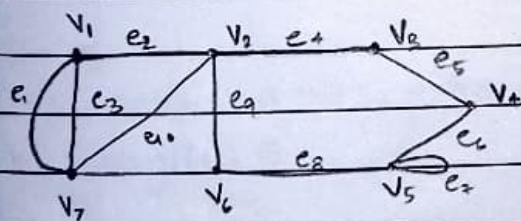
CIRCUIT

- vertices are repeated
- edges are not repeated

- vertices & edges are not repeated

- a path with terminal vertices are the same.

Vertices with which the walk/path/circuit start & end with are terminal vertices.



Walk : $V_1 e_2 V_2 e_{10} V_7 e_3 V_1 e_1 V_7$

Path : $V_1 e_2 V_2 e_4 V_3 e_5 V_4$

Circuit : $V_1 e_2 V_2 e_{10} V_7 e_3 V_1$

* EULER GRAPH

An Euler path that is a circuit is called Euler circuit.
A graph with Eulerian circuit is a eulerian graph.

To determine if graph G has Euler circuit.

1. list degree of vertices.

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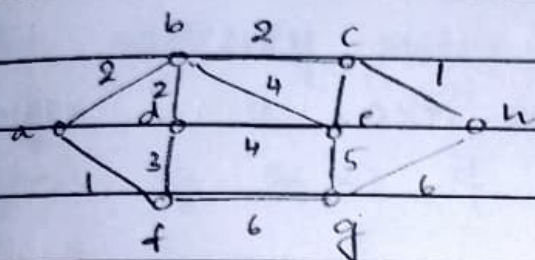
2. if any value is zero, can't have Euler path.
3. if all degrees are even, G has both Euler path & circuit.
4. if exactly 2 vertices are odd, G has E. path but not circuit.

* DIJKSTRA'S ALGORITHM

Find shortest path.

Ex.

1.



	a	b	c	d	e	f	g	h
a	0a	2a	∞	∞	∞	1a	∞	∞
f	-	2a	∞	4f	∞	1a	7f	∞
b	-	2a	4b	4b	6b	-	7f	∞
c	-	-	4b	4b	2c	-	7f	5c
h	-	-	-	-	-	-	5c	

$\therefore$ path : $a \rightarrow b \rightarrow c \rightarrow h$

length : 5

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2.

	A	B	C	D	E	F
A	0A	2A	3A	∞	∞	∞
B	-	2A	3A	∞	3.7B	∞
C	-	-	3A	4.5C	3.7B	∞
E	-	-	-	4.5C	3.7B	7E
D	-	-	-	4.5C	-	6.8D
F	-	-	-	-	-	6.8D

$\therefore$ path : $A \rightarrow C \rightarrow D \rightarrow F$
 length : 6.8 //

* THE HANDSHAKING THEOREM

If $G = (V, E)$ be an undirected graph with e edges.

Then $\sum_{v \in V} \deg_v(v) = 2e$

* THEOREM

The number of vertices of odd degree in a graph is always even.

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* SUBGRAPH

If G & H are two graphs with vertex sets $V(G), V(H)$ and edge sets $E(G), E(H)$. If $V(H) \subseteq V(G)$ & $E(H) \subseteq E(G)$ then H is a sub-graph of G .

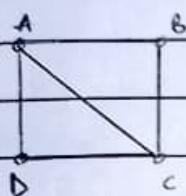
* SPANNING SUBGRAPH

A spanning subgraph contains all vertices of G .

* ADJACENCY MATRIX

The adjacency matrix of a graph G with n vertices & no self edges is a $n \times n$ matrix $A = \{a_{ij}\}$ whose elements are given by $a_{ij} = 1$ if there is an edge b/w i th & j th vertices.

eg:

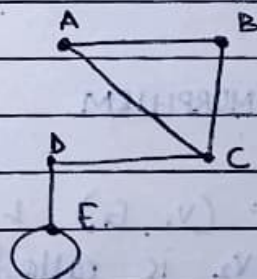


$$\begin{array}{c}
 \begin{array}{cc} & \begin{array}{cccc} & A & B & C & D \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \end{array} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}
 \end{array}
 \end{array}$$

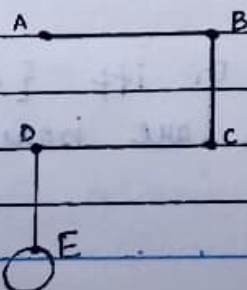
Ex 10

1.

$$\begin{array}{c}
 \begin{array}{cc} & \begin{array}{ccccc} & A & B & C & D & E \end{array} \\ \begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}
 \end{array}
 \end{array}$$



graphs :

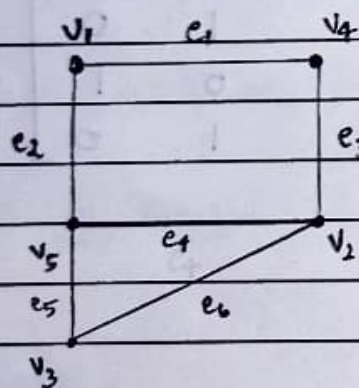


* INCIDENCE MATRIX

If $G = (V, E)$ is a graph with n vertices & m edges all labelled. The incidence matrix $B = \{b_{ij}\}$ is a $n \times m$ matrix where $b_{ij} = 1$ when e_j is incident on v_i .

eg:

	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	1	0	0	0	0
v_2	0	0	1	1	0	1
v_3	0	0	0	0	1	1
v_4	1	0	1	0	0	0
v_5	0	1	0	1	1	0

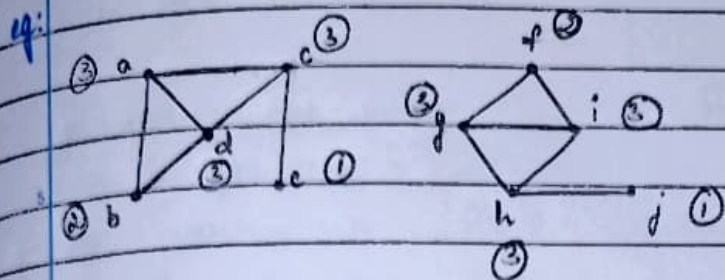


* GRAPH ISOMORPHISM

Let $G_1 = (V_1, E_1)$ & $G_2 = (V_2, E_2)$ be graphs. A function $f: V_1 \rightarrow V_2$ is called graph isomorphism if,

(i) f is one-one & onto

(ii) for all $a, b \in V_1$, $\{a, b\} \in E_1$ iff $\{f(a), f(b)\} \in E_2$ when such a fn. exists, G_1 & G_2 are isomorphic: $G_1 \cong G_2$.



1. each graph has $|V| = 5$, $|E| = 6$

2. there are
 1-1 deg
 1-2 deg
 3-3 deg vertices

3. vertex correspondance

a - g
 b - f
 c - h
 d - i
 e - j

$$A(G_2) =$$

	g	f	h	i	j
g	0	1	1	1	0
f	1	0	0	1	0
h	1	0	0	1	1
i	1	1	1	0	0
j	0	0	1	0	0

4. adjacency matrix

$$A(G_1) =$$

	a	b	c	d	e
a	0	1	1	1	0
b	1	0	0	1	0
c	1	0	0	1	1
d	1	1	1	0	0
e	0	0	1	0	0

$\therefore A(G_1) \cong A(G_2)$, hence G_1 & G_2 are isomorphic.